

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

P R O J E C T R E P O R T

Submitted by

SIVA KARTHICK

25ITR137

JAVA ASSIGNMENT

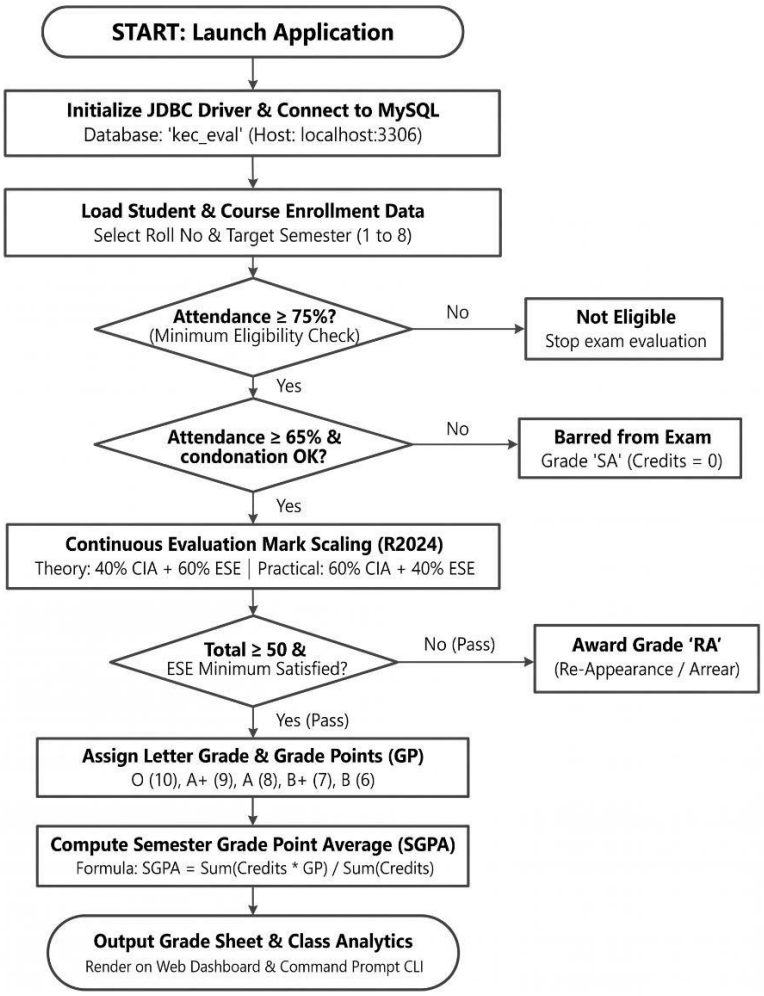
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

FLOW CHART:



PROCEDURE:

- Load the MySQL Connector/J driver to enable communication between the Java application and the MySQL database.

- Establish a connection to the kec_eval database running on the local MySQL server using DriverManager.getConnection().

- Store and retrieve student details, including roll number, name, department, and semester, along with course information such as course code, title, credits, and course type.

- Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.

- Verify examination eligibility based on attendance percentage. Students with 75% or above are eligible, those with 65%–74% require medical condonation verification, and those below 65% are marked as Shortage of Attendance (SA).

- Record CIA-1, CIA-2, and assignment marks, and calculate the scaled internal assessment marks according to the specified theory and practical course weightages.

- Record end-semester examination marks out of 100, verify the minimum passing requirements, and scale the marks according to the applicable theory and practical .

- Calculate the total marks by combining scaled CIA and ESE marks. Assign letter grades and grade points according to the specified grading scale, marking students as RA if they fail the examination or score below the required passing mark.

- Generate student grade sheets, attendance shortage reports, and department-wise pass percentage reports. Display the results through the interactive web dashboard and the command prompt console**.**

DATABASE SCHEMA:

1.STUDENTS TABLE:

```
mysql> describe students;
+------------------+--------------+------+-----+-------------------+-------------------+
| Field            | Type         | Null | Key | Default           | Extra             |
+------------------+--------------+------+-----+-------------------+-------------------+
| roll_no          | varchar(20)  | NO   | PRI | NULL              |                   |
| name             | varchar(100) | NO   |     | NULL              |                   |
| department       | varchar(20)  | NO   |     | NULL              |                   |
| current_semester | int          | NO   |     | 3                 |                   |
| created_at       | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------------+--------------+------+-----+-------------------+-------------------+
5 rows in set (0.03 sec)
```

2. COURSES TABLE:

```
mysql> describe courses;
+-------------+-----------------------------------------+------+-----+-------------------+-------------------+
| Field       | Type                                    | Null | Key | Default           | Extra             |
+-------------+-----------------------------------------+------+-----+-------------------+-------------------+
| course_code | varchar(20)                             | NO   | PRI | NULL              |                   |
| title       | varchar(150)                            | NO   |     | NULL              |                   |
| credits     | int                                     | NO   |     | NULL              |                   |
| course_type | enum('THEORY','PRACTICAL','EMBEDDED')   | NO   |     | NULL              |                   |
| semester    | int                                     | NO   |     | 3                 |                   |
| created_at  | timestamp                               | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-------------+-----------------------------------------+------+-----+-------------------+-------------------+
6 rows in set (0.01 sec)
```

3. ENROLLMENTS TABLE:

```
mysql> describe enrollments;
+------------------+-------------+------+-----+-------------------+-------------------+
| Field            | Type        | Null | Key | Default           | Extra             |
+------------------+-------------+------+-----+-------------------+-------------------+
| enrollment_id    | int         | NO   | PRI | NULL              | auto_increment    |
| roll_no          | varchar(20) | NO   | MUL | NULL              |                   |
| course_code      | varchar(20) | NO   | MUL | NULL              |                   |
| semester         | int         | NO   |     | 3                 |                   |
| total_classes    | int         | NO   |     | 0                 |                   |
| attended_classes | int         | NO   |     | 0                 |                   |
| is_condoned      | tinyint(1)  | NO   |     | 0                 |                   |
| created_at       | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------------+-------------+------+-----+-------------------+-------------------+
8 rows in set (0.01 sec)
```

4. ASSESSMENT MARKS TABLE:

```
mysql> describe assessment_marks;
+-----------------+-------------+------+-----+-------------------+-----------------------------------------------+
| Field           | Type        | Null | Key | Default           | Extra                                         |
+-----------------+-------------+------+-----+-------------------+-----------------------------------------------+
| mark_id         | int         | NO   | PRI | NULL              | auto_increment                                |
| enrollment_id   | int         | NO   | UNI | NULL              |                                               |
| cia1_marks      | decimal(5,2) | YES  |     | NULL              |                                               |
| cia2_marks      | decimal(5,2) | YES  |     | NULL              |                                               |
| assignment_marks | decimal(5,2) | YES  |     | NULL              |                                               |
| ese_marks       | decimal(5,2) | YES  |     | NULL              |                                               |
| is_ese_absent   | tinyint(1)  | NO   |     | 0                 |                                               |
| updated_at      | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED on update CURRENT_TIMESTAMP |
+-----------------+-------------+------+-----+-------------------+-----------------------------------------------+
8 rows in set (0.02 sec)
```

SOURCE CODE:

MAINAPP.JAVA:

```
package com.kec.academic;

import java.sql.SQLException;
import java.util.Scanner;

public class MainApp {

    private static final Scanner sc = new Scanner(System.in);

    private static final StudentDAO studentDAO = new StudentDAO();
    private static final CourseDAO courseDAO = new CourseDAO();
    private static final EnrollmentDAO enrollmentDAO = new EnrollmentDAO();
    private static final AttendanceDAO attendanceDAO = new AttendanceDAO();
    private static final AssessmentDAO assessmentDAO = new AssessmentDAO();
    private static final GradeDAO gradeDAO = new GradeDAO();
    private static final GradeResultDAO gradeResultDAO = new GradeResultDAO();
    private static final SGPACalculator sgpaCalculator = new SGPACalculator();
    private static final FinalGradeService finalGradeService = new FinalGradeService();
    private static final AssessmentWeightageDAO weightageDAO = new AssessmentWeightageDAO();
    private static final AssessmentWeightageMenu weightageMenu = new AssessmentWeightageMenu();

    public static void main(String[] args) {
        System.out.println("==========================================");
        System.out.println(" KEC ACADEMIC PERFORMANCE MANAGEMENT SYSTEM");
        System.out.println("==========================================");

        while (true) {
            showMainMenu();
            String choice = ask("Enter your choice: ");

            try {
                switch (choice) {
                    case "1": studentMenu(); break;
                    case "2": courseMenu(); break;
                    case "3": enrollmentMenu(); break;
                    case "4": attendanceMenu(); break;
                    case "5": assessmentMenu(); break;
                    case "6": gradeReportMenu(); break;
                    case "7": calculateSGPA(); break;
```

```
case "8": weightageMenu.showMenu(sc); break;
                case "0":

    System.out.println("\nThank you for using KEC Academic System!");
                    sc.close();
                    return;
                default:
                    System.out.println("Invalid choice!");
            }
        } catch (SQLException e) {
            System.out.println("\nDatabase operation failed: " + e.getMessage());
            System.out.println("Check your database connection and table structure.");
        } catch (Exception e) {
            System.out.println("\nOperation failed: " + e.getMessage());
        }
    }
}

private static void showMainMenu() {
    System.out.println("\n==========================================");
    System.out.println("              MAIN MENU");
    System.out.println("==========================================");
    System.out.println("1. Student Management");
    System.out.println("2. Course Management");
    System.out.println("3. Enrollment Management");
    System.out.println("4. Attendance Management");
    System.out.println("5. Assessment Management");
    System.out.println("6. View Grade Report");
    System.out.println("7. Calculate SGPA");
    System.out.println("8. Assessment Weightage Configuration");
    System.out.println("0. Exit");
    System.out.println("==========================================");
}

// ---------- Student ----------

private static void studentMenu() throws SQLException {
    while (true) {
        System.out.println("\n---------- STUDENT MANAGEMENT ----------");
        System.out.println("1. Add Student");
        System.out.println("2. View All Students");
        System.out.println("3. Search Student");
        System.out.println("4. Update Student");
        System.out.println("5. Delete Student");
        System.out.println("0. Back to Main Menu");

        switch (ask("Enter choice: ")) {
            case "1": addStudent(); break;
            case "2": studentDAO.viewAllStudents(); break;
            case "3": studentDAO.searchStudent(askUpper("Enter roll number: ")); break;
            case "4": updateStudent(); break;
            case "5": deleteStudent(); break;
            case "0": return;
            default: System.out.println("Invalid choice!");
        }
    }
}

private static void addStudent() throws SQLException {
    String rollNo = askUpper("Enter roll number: ");
    String name = ask("Enter student name: ");
    String dept = askUpper("Enter department: ");
    int sem = readInt("Enter current semester: ");
```

```
 if (rollNo.isEmpty() || name.isEmpty() || dept.isEmpty()) {
            System.out.println("Student details cannot be empty.");
            return;
        }

    if (sem < 1 || sem > 8) {
            System.out.println("Semester must be between 1 and 8.");
            return;
        }
        studentDAO.addStudent(rollNo, name, dept, sem);
    }

    private static void updateStudent() throws SQLException {
        String rollNo = askUpper("Enter roll number of student to update: ");
        String name = ask("Enter new name: ");
        String dept = askUpper("Enter new department: ");
        int sem = readInt("Enter new current semester: ");

        if (rollNo.isEmpty() || name.isEmpty() || dept.isEmpty() || sem < 1 || sem > 8) {
            System.out.println("Invalid student details.");
            return;
        }
        studentDAO.updateStudent(rollNo, name, dept, sem);
    }

    private static void deleteStudent() throws SQLException {
        String rollNo = askUpper("Enter roll number to delete: ");
        if (rollNo.isEmpty()) {
            System.out.println("Roll number cannot be empty.");
            return;
        }

        String confirm = ask("Are you sure you want to delete this student? (yes/no): ");
        if (confirm.equalsIgnoreCase("yes")) {
            studentDAO.deleteStudent(rollNo);
        } else {
            System.out.println("Deletion cancelled.");
        }
    }

    // ---------- Course ----------

    private static void courseMenu() throws SQLException {
        while (true) {
            System.out.println("\n---------- COURSE MANAGEMENT ----------");
            System.out.println("1. Add Course");
            System.out.println("2. View All Courses");
            System.out.println("3. Search Course");
            System.out.println("0. Back to Main Menu");

            switch (ask("Enter choice: ")) {
                case "1": addCourse(); break;
                case "2": courseDAO.viewAllCourses(); break;
                case "3": courseDAO.searchCourse(askUpper("Enter course code: ")); break;
                case "0": return;
                default: System.out.println("Invalid choice!");
            }
        }
    }

    private static void addCourse() throws SQLException {
        String code = askUpper("Enter course code: ");
        String title = ask("Enter course title: ");
        int credits = readInt("Enter course credits: ");

        System.out.println("Course Types: THEORY, PRACTICAL, EMBEDDED");
```

```
        String type = askUpper("Enter course type: ");

        if (code.isEmpty() || title.isEmpty() || credits <= 0) {


    System.out.println("Invalid course details.");
            return;
        }
        if (!type.equals("THEORY") && !type.equals("PRACTICAL") && !type.equals("EMBEDDED")) {
            System.out.println("Invalid course type.");
            return;
        }
        courseDAO.addCourse(code, title, credits, type);
    }

    // ---------- Enrollment ----------

    private static void enrollmentMenu() throws SQLException {
        while (true) {
            System.out.println("\n---------- ENROLLMENT MANAGEMENT ----------");
            System.out.println("1. Enroll Student in Course");
            System.out.println("2. View Enrollment Records");
            System.out.println("0. Back to Main Menu");

            switch (ask("Enter choice: ")) {
                case "1": enrollStudent(); break;
                case "2": enrollmentDAO.viewEnrollments(); break;
                case "0": return;
                default: System.out.println("Invalid choice!");
            }
        }
    }

    private static void enrollStudent() throws SQLException {
        String rollNo = askUpper("Enter student roll number: ");
        String courseCode = askUpper("Enter course code: ");
        int sem = readInt("Enter semester: ");

        if (rollNo.isEmpty() || courseCode.isEmpty() || sem < 1 || sem > 8) {
            System.out.println("Invalid enrollment details.");
            return;
        }
        enrollmentDAO.enrollStudent(rollNo, courseCode, sem);
    }

    // ---------- Attendance ----------

    private static void attendanceMenu() throws SQLException {
        while (true) {
            System.out.println("\n---------- ATTENDANCE MANAGEMENT ----------");
            System.out.println("1. Update Attendance");
            System.out.println("2. Update Medical Condonation");
            System.out.println("3. View Attendance Report");
            System.out.println("0. Back to Main Menu");

            switch (ask("Enter choice: ")) {
                case "1": updateAttendance(); break;
                case "2": updateCondonation(); break;
                case "3": attendanceDAO.viewAttendanceReport(); break;
                case "0": return;
                default: System.out.println("Invalid choice!");
            }
        }
    }
```

```
private static void updateAttendance() throws SQLException {
    int enrollmentId = readInt("Enter enrollment ID: ");
    int total = readInt("Enter total classes conducted: ");
    int attended = readInt("Enter classes attended: ");

    if (enrollmentId <= 0 || total < 0 || attended < 0 || attended > total) {
        System.out.println("Invalid attendance details.");
        return;
    }
    attendanceDAO.updateAttendance(enrollmentId, total, attended);
}

private static void updateCondonation() throws SQLException {
    int enrollmentId = readInt("Enter enrollment ID: ");
    if (enrollmentId <= 0) {
        System.out.println("Invalid enrollment ID.");
        return;
    }

    String answer = ask("Has medical condonation been approved? (yes/no): ");
    if (!isYesOrNo(answer)) {
        System.out.println("Please enter yes or no.");
        return;
    }
    attendanceDAO.updateCondonation(enrollmentId, answer.equalsIgnoreCase("yes"));
}

// ---------- Assessment ----------

private static void assessmentMenu() throws SQLException {
    while (true) {
        System.out.println("\n---------- ASSESSMENT MANAGEMENT ----------");
        System.out.println("1. Enter or Update Assessment Marks");
        System.out.println("2. View Assessment Records");
        System.out.println("0. Back to Main Menu");

        switch (ask("Enter choice: ")) {
            case "1": enterMarks(); break;
            case "2": assessmentDAO.viewAssessments(); break;
            case "0": return;
            default: System.out.println("Invalid choice!");
        }
    }
}

private static void enterMarks() throws SQLException {
    int enrollmentId = readInt("Enter enrollment ID: ");
    if (enrollmentId <= 0) {
        System.out.println("Enrollment ID must be positive.");
        return;
    }

    System.out.println("Enter marks from 0 to 100. Leave blank if not entered.");
    Double cia1 = readMark("CIA 1 marks: ");
    Double cia2 = readMark("CIA 2 marks: ");
    Double cia3 = readMark("CIA 3 marks: ");
    Double assignment = readMark("Assignment marks: ");
    Double quiz = readMark("Quiz marks: ");
    Double lab = readMark("Lab marks: ");
    Double ese = readMark("ESE marks: ");

    String answer = ask("Was the student absent for ESE? (yes/no): ");
    if (!isYesOrNo(answer)) {
```

```
System.out.println("Please enter yes or no.");
            return;
        }
        boolean absent = answer.equalsIgnoreCase("yes");

        Double[] all = {cia1, cia2, cia3, assignment, quiz, lab, ese};
        for (Double m : all) {
            if (!validMark(m)) {
                System.out.println("Marks must be numbers between 0 and 100.");
                return;
            }
        }

        assessmentDAO.saveAssessment(enrollmentId, cia1, cia2, cia3,
                assignment, quiz, lab, ese, absent);
    }

    private static Double readMark(String prompt) {
        String input = ask(prompt);
        if (input.isEmpty()) {
            return null;
        }
        try {
            return Double.parseDouble(input);
        } catch (NumberFormatException e) {
            return Double.NaN; // caught later by validMark
        }
    }

    private static boolean validMark(Double m) {
        return m == null || (Double.isFinite(m) && m >= 0 && m <= 100);
    }

    // ---------- Grade report ----------

    private static void gradeReportMenu() throws SQLException {
        while (true) {
            System.out.println("\n---------- GRADE REPORT ----------");
            System.out.println("1. View Student Grade Details");
            System.out.println("2. View Saved Final Grades");
            System.out.println("3. Calculate and Save Final Grades");
            System.out.println("0. Back to Main Menu");

            String choice = ask("Enter choice: ");
            if (choice.equals("0")) {
                return;
            }
            if (!choice.equals("1") && !choice.equals("2") && !choice.equals("3")) {
                System.out.println("Invalid choice!");
                continue;
            }

            String rollNo = askUpper("Enter student roll number: ");
            if (rollNo.isEmpty()) {
                System.out.println("Roll number cannot be empty.");
                continue;
            }

            if (choice.equals("1")) {
                gradeDAO.viewStudentGradeDetails(rollNo);
            } else if (choice.equals("2")) {
                gradeResultDAO.viewGrades(rollNo);
```

```
        } else {
                finalGradeService.calculateGradesForStudent(rollNo);
            }
        }
    }

    // ---------- SGPA ----------

    private static void calculateSGPA() throws SQLException {
        System.out.println("\n---------- SGPA CALCULATOR ----------");
        String rollNo = askUpper("Enter student roll number: ");
        int sem = readInt("Enter semester: ");

        if (rollNo.isEmpty()) {
            System.out.println("Roll number cannot be empty.");
            return;
        }
        if (sem < 1 || sem > 8) {
            System.out.println("Semester must be between 1 and 8.");
            return;
        }
        sgpaCalculator.calculateSGPA(rollNo, sem);
    }

    // ---------- Input helpers ----------

    private static String ask(String prompt) {
        System.out.print(prompt);
        return sc.nextLine().trim();
    }

    private static String askUpper(String prompt) {
        return ask(prompt).toUpperCase();
    }

    private static boolean isYesOrNo(String s) {
        return s.equalsIgnoreCase("yes") || s.equalsIgnoreCase("no");
    }

    private static int readInt(String prompt) {
        while (true) {
            try {
                return Integer.parseInt(ask(prompt));
            } catch (NumberFormatException e) {
                System.out.println("Please enter a valid whole number.");
            }
        }
    }
}
```

DB CONNECTION.JAVA:

```
package com.kec.academic;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {

    private static final String URL = "jdbc:mysql://localhost:3306/kec_academics";
    private static final String USER = "root";

    public static Connection getConnection() throws SQLException {
        String password = System.getenv("KEC_DB_PASSWORD");

        if (password == null || password.isBlank()) {
            throw new SQLException("Database password not found. Set KEC_DB_PASSWORD first.");
        }

        return DriverManager.getConnection(URL, USER, password);
    }
}
```

OUTPUT SCREENSHOTS:

MAIN MENU:

```
-------------------------------- MAIN MENU -----------------------------------------
 1. Student Registration (Add New IT Student)
 2. Course Catalog Management (Semester 3 Courses)
 3. Course Enrollment & Attendance Tracking
 4. Incremental Marks Entry (Phase 1: CIA1/2 | Phase 2: Asg/Lab | Phase 3: ESE)
 5. Generate Student Semester Mark Sheet (SGPA & Equivalent %)
 6. Generate Attendance Defaulter & Condonation Report
 7. Generate Class Performance & Result Summary
 8. Run Full Automated Demonstration (Proof of Execution)
 9. Launch Interactive Web Dashboard (Browser GUI @ http://localhost:8080)
 0. Exit
-------------------------------------------------------------------------------------
Enter your choice (1-8): 1
```

STUDENT & COURSE MANAGEMENT:

1.STUDENT REGISTRATION:

```
--- [1] Student Registration ---
Enter Roll Number (e.g., 24ITR050): 25ITL187
Enter Student Full Name: Lingesh B
Enter Department [Press Enter for IT]:
Enter Current Semester [Press Enter for 3]:
[SUCCESS] Student 25ITL187 (Lingesh B) registered successfully.

Press [Enter] to return to the Main Menu...
```

2. COURSE CATALOG:

```
--- [2] Course Catalog Management ---
Enter Course Code (e.g., 24ITR31 / 24ITT32): 24ITT11
Enter Course Title: Game Development
Enter Credits (e.g., 2, 3, 4): 2
Enter Course Type (1: THEORY, 2: PRACTICAL, 3: EMBEDDED): 2
Enter Offering Semester (1-8): 4
[SUCCESS] Course 24ITT11 [PRACTICAL, 2 Credits] added successfully.

Press [Enter] to return to the Main Menu...
```

3. COURSE ENROLLMENT & ATTENDANCE TRACKING:

```
--- [3] Course Enrollment & Attendance Tracking ---
Enter Student Roll Number: 25ITL187
Enter Course Code: 24ITC31
Enter Semester: 3
Enter Total Classes Held: 60
Enter Classes Attended: 52
Calculated Attendance: 86.67%
[INFO] Attendance >= 75%. Eligible for ESE.
[SUCCESS] Enrollment and Attendance updated. (Enrollment ID: 413)

Press [Enter] to return to the Main Menu...
```

INCREMENTAL MARKS ENTRY:

1.PHASE 1 (MID-SEMESTER):

```
----------------- INCREMENTAL MARKS ENTRY (THEN AND NOW) -----------------
 Select Assessment Phase:
 1. Phase 1 (Mid-Semester): CIA Test 1 (0-100) & CIA Test 2 (0-100)
 2. Phase 2 (End of Semester - CIA): Assignments / Quizzes / Lab (0-100)
 3. Phase 3 (Post Semester): End Semester Exam (0-100) & Absentee (AB)
Enter Phase (1-3): 1
Enter Student Roll Number: 25ITL187
Enter Course Code: 24ITC31
Enter Semester: 3
Enter CIA Test 1 Marks (0-100): 92
Enter CIA Test 2 Marks (0-100): 93
[SUCCESS] Phase 1 Marks recorded successfully.

Press [Enter] to return to the Main Menu...
```

2.PHASE 2 (END OF SEMESTER - CIA):

```
----------------- INCREMENTAL MARKS ENTRY (THEN AND NOW) -----------------
 Select Assessment Phase:
 1. Phase 1 (Mid-Semester): CIA Test 1 (0-100) & CIA Test 2 (0-100)
 2. Phase 2 (End of Semester - CIA): Assignments / Quizzes / Lab (0-100)
 3. Phase 3 (Post Semester): End Semester Exam (0-100) & Absentee (AB)
Enter Phase (1-3): 2
Enter Student Roll Number: 25ITL187
Enter Course Code: 24ITC31
Enter Semester: 3
Enter Assignment / Quiz / Lab Evaluation Marks (0-100): 90
[SUCCESS] Phase 2 Marks recorded successfully.

Press [Enter] to return to the Main Menu...
```

3. PHASE 3 (POST SEMESTER):

```
---------------- INCREMENTAL MARKS ENTRY (THEN AND NOW) ----------------
 Select Assessment Phase:
 1. Phase 1 (Mid-Semester): CIA Test 1 (0-100) & CIA Test 2 (0-100)
 2. Phase 2 (End of Semester - CIA): Assignments / Quizzes / Lab (0-100)
 3. Phase 3 (Post Semester): End Semester Exam (0-100) & Absentee (AB)
Enter Phase (1-3): 3
Enter Student Roll Number: 25ITL187
Enter Course Code: 24ITC31
Enter Semester: 3
Is student Absent (AB) in ESE? (y/n): n
Enter End Semester Exam (ESE) Marks (0-100): 87
[SUCCESS] Phase 3 ESE Marks recorded successfully.

Press [Enter] to return to the Main Menu...
```

ATTENDANCE & GRADING RULES:

1.ATTENDANCE FILTER:

```
--------------------------------------------------------------------------------
Enter your choice (1-8): 6

Enter Semester to check (e.g., 3): 3

================================================================================================================
                    KONGU ENGINEERING COLLEGE (AUTONOMOUS) - REGULATION 2024
                         ATTENDANCE DEFAULTER & CONDONATION REPORT
                                  SEMESTER: 3
================================================================================================================
Roll No      Student Name        Dept     Course     Held     Attended Attd %     Condonation  Eligibility / Grade
----------------------------------------------------------------------------------------------------------------
24ADR010     Ananya S            AIDS     24ADT31    60       32          53.3% NOT APPROVED BARRED (Grade SA)
24ADR010     Ananya S            AIDS     24ADC31    60       32          53.3% NOT APPROVED BARRED (Grade SA)
24ADR010     Ananya S            AIDS     24ADL31    30       16          53.3% NOT APPROVED BARRED (Grade SA)
24CER003     Aravind K           CIVIL    24CET31    60       42          70.0% APPROVED     CONDONED (Eligible)
24CER003     Aravind K           CIVIL    24CEC31    60       42          70.0% APPROVED     CONDONED (Eligible)
24CER003     Aravind K           CIVIL    24CEL31    30       21          70.0% NOT APPROVED PENDING (Grade SA)
----------------------------------------------------------------------------------------------------------------
 Note: <65% Attendance: Shortage of Attendance (SA - Barred) | 65-74.9%: Requires Medical Condonation
================================================================================================================


Press [Enter] to return to the Main Menu...
```

STUDENT SEMESTER MARK SHEET:

```
Enter Student Roll Number (e.g., 24ITR001): 24ITR099
Enter Semester (e.g., 3): 3

================================================================================================================
                                KONGU ENGINEERING COLLEGE (AUTONOMOUS)
                                    PERUNDURAI, ERODE - 638060
                              OFFICE OF THE CONTROLLER OF EXAMINATIONS
                            GRADE SHEET - REGULATION 2024 (AUTONOMOUS)
================================================================================================================
 Roll Number : 24ITR099          Student Name : Murasoli B                     Semester : 3
 Department  : IT                Regulation   : R2024 (KEC)                    Date     : 20-10-2026
----------------------------------------------------------------------------------------------------------------
Course    Course Title               Type      Cr   Attd%    CIA1   CIA2   Asg    Sc.CIA   ESE(100) Sc.ESE  Total  Grd  Result
----------------------------------------------------------------------------------------------------------------
24ITC31   Full Stack Development      EMBEDDED  4      96.7% 94.0   96.0   95.0      47.50 94.0     47.0     94.50 O    PASS
24ITL31   Programming in Java Labo...  PRACTICAL 1     100.0% 98.0   98.0   100.0     59.16 96.0     38.4     97.56 O    PASS
24ITT32   Programming in Java         THEORY    3      97.8% 90.0   92.0   94.0      36.76 90.0     54.0     90.76 O    PASS
----------------------------------------------------------------------------------------------------------------
 Total Enrolled Credits : 8      |   Credits Earned : 8      |   Weighted Grade Points : 80.00
 SEMESTER GRADE POINT AVERAGE (SGPA) : 10.00 / 10.00
 EQUIVALENT PERCENTAGE               : 100.00%
 SEMESTER EVALUATION OUTCOME        : [PASSED] ALL REGISTERED COURSES CLEARED
================================================================================================================
 Grading Standard (R2024): O(91-100)->10 | A+(81-90)->9 | A(71-80)->8 | B+(61-70)->7 | B(50-60)->6 | RA/SA/AB->0
 Rules: Theory (40 CIA/60 ESE, ESE Min 27/60) | Practical (60 CIA/40 ESE, ESE Min 20/40) | Pass Minimum: 50 Marks
================================================================================================================


Press [Enter] to return to the Main Menu...
```

DEPARTMENT WISE CLASS PERFORMANCE & RESULT SUMMARY:

```
Enter your choice (1-8): 7

Enter Semester (e.g., 3): 3

================================================================================================
                    KONGU ENGINEERING COLLEGE (AUTONOMOUS) - REGULATION 2024
                     DEPARTMENT-WISE CLASS PERFORMANCE & RESULT SUMMARY
                                      SEMESTER: 3
================================================================================================
Department    Total Students    Passed All     With Arrears    Pass Rate (%)   Class Avg SGPA
------------------------------------------------------------------------------------------------
AIDS          1                 0              1                     0.00%            0.00
CHEM          1                 1              0                   100.00%            9.11
CIVIL         1                 0              1                     0.00%            8.44
CSE           2                 2              0                   100.00%            8.42
ECE           1                 1              0                   100.00%            9.00
EEE           1                 1              0                   100.00%            9.00
IT            5                 5              0                   100.00%            8.47
MECH          1                 0              1                     0.00%            6.58
================================================================================================


Press [Enter] to return to the Main Menu...
```

DATABASE:

STUDENTS DATABASE:

```
mysql> USE kec_eval;
Database changed
mysql> SELECT * FROM students;
+----------+--------------------+------------+------------------+---------------------+
| roll_no  | name               | department | current_semester | created_at          |
+----------+--------------------+------------+------------------+---------------------+
| 22ADR071 | Sudharsan G        | AIDS       |                7 | 2026-10-19 21:29:39 |
| 22ADR081 | Akilesh R          | AIDS       |                8 | 2026-10-19 21:29:39 |
| 22CER071 | Saranya V          | CIVIL      |                7 | 2026-10-19 21:29:39 |
| 22CER081 | Yogeshwaran T      | CIVIL      |                8 | 2026-10-19 21:29:39 |
| 22CHR071 | Swetha T           | CHEM       |                7 | 2026-10-19 21:29:39 |
| 22CHR081 | Balamurugan S      | CHEM       |                8 | 2026-10-19 21:29:39 |
| 22CSR071 | Praveen Kumar R    | CSE        |                7 | 2026-10-19 21:29:39 |
| 22CSR072 | Priyanga M         | CSE        |                7 | 2026-10-19 21:29:39 |
| 22CSR081 | Venkatesh S        | CSE        |                8 | 2026-10-19 21:29:39 |
| 22CSR082 | Vidhya R           | CSE        |                8 | 2026-10-19 21:29:39 |
| 22ECR071 | Rajesh K           | ECE        |                7 | 2026-10-19 21:29:39 |
| 22ECR081 | Vigneshwar B       | ECE        |                8 | 2026-10-19 21:29:39 |
| 22EER071 | Rithika S          | EEE        |                7 | 2026-10-19 21:29:39 |
| 22EER081 | Vishnu Vardhan M   | EEE        |                8 | 2026-10-19 21:29:39 |
| 22ITR071 | Manikandan P       | IT         |                7 | 2026-10-19 21:29:39 |
| 22ITR072 | Nandhini S         | IT         |                7 | 2026-10-19 21:29:39 |
```

COURSES DATABASE:

```
mysql> SELECT * FROM courses;
+-------------+-----------------------------------------------+---------+-------------+----------+---------------------+
| course_code | title                                         | credits | course_type | semester | created_at          |
+-------------+-----------------------------------------------+---------+-------------+----------+---------------------+
| 24ADC31     | Artificial Intelligence & Knowledge Rep.      |       4 | EMBEDDED    |        3 | 2026-10-19 20:59:10 |
| 24ADC41     | Machine Learning Algorithms & Systems         |       4 | EMBEDDED    |        4 | 2026-10-19 20:59:10 |
| 24ADC51     | Natural Language Processing (NLP)             |       4 | EMBEDDED    |        5 | 2026-10-19 20:59:10 |
| 24ADC61     | Generative AI & Large Language Models         |       4 | EMBEDDED    |        6 | 2026-10-19 20:59:10 |
| 24ADC71     | Reinforcement Learning & Autonomous Agents    |       4 | EMBEDDED    |        7 | 2026-10-19 20:59:10 |
| 24ADL31     | Python for AI & Data Science Laboratory       |       1 | PRACTICAL   |        3 | 2026-10-19 20:59:10 |
| 24ADL41     | Machine Learning & Data Mining Lab            |       1 | PRACTICAL   |        4 | 2026-10-19 20:59:10 |
| 24ADL51     | Deep Learning & NLP Laboratory                |       1 | PRACTICAL   |        5 | 2026-10-19 20:59:10 |
| 24ADL61     | Computer Vision & GenAI Laboratory            |       1 | PRACTICAL   |        6 | 2026-10-19 20:59:10 |
| 24ADP71     | AI Capstone Project Phase I                   |       3 | PRACTICAL   |        7 | 2026-10-19 20:59:10 |
| 24ADP81     | AI Capstone Project Phase II / Industry Track |      10 | PRACTICAL   |        8 | 2026-10-19 20:59:10 |
| 24ADT31     | Mathematical Foundations for Data Science     |       4 | THEORY      |        3 | 2026-10-19 20:59:10 |
| 24ADT41     | Statistical Learning & Data Mining            |       4 | THEORY      |        4 | 2026-10-19 20:59:10 |
| 24ADT51     | Deep Learning & Neural Networks               |       4 | THEORY      |        5 | 2026-10-19 20:59:10 |
| 24ADT61     | Computer Vision & Video Analytics             |       4 | THEORY      |        6 | 2026-10-19 20:59:10 |
| 24ADT71     | AI Ethics, Governance & MLOps Infrastructure  |       3 | THEORY      |        7 | 2026-10-19 20:59:10 |
| 24CEC31     | Surveying and Geomatics                       |       4 | EMBEDDED    |        3 | 2026-10-19 20:59:10 |
| 24CEC41     | Fluid Mechanics & Hydraulic Machinery         |       4 | EMBEDDED    |        4 | 2026-10-19 20:59:10 |
| 24CEC51     | Soil Mechanics & Geotechnical Engineering     |       4 | EMBEDDED    |        5 | 2026-10-19 20:59:10 |
| 24CEC61     | Environmental Engineering & Water Treatment   |       4 | EMBEDDED    |        6 | 2026-10-19 20:59:10 |
| 24CEC71     | Transportation Engineering & GIS              |       4 | EMBEDDED    |        7 | 2026-10-19 20:59:10 |
| 24CEL31     | Surveying Field Work Laboratory               |       1 | PRACTICAL   |        3 | 2026-10-19 20:59:10 |
| 24CEL41     | Materials Testing & Hydraulics Lab            |       1 | PRACTICAL   |        4 | 2026-10-19 20:59:10 |
| 24CEL51     | Geotechnical Engineering Lab                  |       1 | PRACTICAL   |        5 | 2026-10-19 20:59:10 |
| 24CEL61     | Environmental Engineering Laboratory          |       1 | PRACTICAL   |        6 | 2026-10-19 20:59:10 |
| 24CEP71     | Project Work Phase I                          |       3 | PRACTICAL   |        7 | 2026-10-19 20:59:10 |
```

ENROLLMENTS DATABASE:

```
mysql> SELECT * FROM enrollments;
+---------------+----------+-------------+----------+---------------+------------------+-------------+---------------------+
| enrollment_id | roll_no  | course_code | semester | total_classes | attended_classes | is_condoned | created_at          |
+---------------+----------+-------------+----------+---------------+------------------+-------------+---------------------+
|             1 | 24ITR001 | 24ITT31     |        3 |            60 |               49 |           0 | 2026-10-19 20:59:10 |
|             2 | 24ITR001 | 24ITC31     |        3 |            60 |               49 |           0 | 2026-10-19 20:59:10 |
|             3 | 24ITR001 | 24ITT32     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
|             4 | 24ITR001 | 24ITC32     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
|             5 | 24ITR001 | 24ITT33     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|             6 | 24ITR001 | 24MNT31     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|             7 | 24ITR001 | 24ITL31     |        3 |            30 |               26 |           0 | 2026-10-19 20:59:10 |
|             8 | 24ITR001 | 24ITL32     |        3 |            30 |               27 |           0 | 2026-10-19 20:59:10 |
|             9 | 24ITR001 | 24ITL33     |        3 |            30 |               28 |           0 | 2026-10-19 20:59:10 |
|            10 | 24ITR002 | 24ITT31     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
|            11 | 24ITR002 | 24ITC31     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
|            12 | 24ITR002 | 24ITT32     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|            13 | 24ITR002 | 24ITC32     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|            14 | 24ITR002 | 24ITT33     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|            15 | 24ITR002 | 24MNT31     |        3 |            60 |               56 |           0 | 2026-10-19 20:59:10 |
|            16 | 24ITR002 | 24ITL31     |        3 |            30 |               27 |           0 | 2026-10-19 20:59:10 |
|            17 | 24ITR002 | 24ITL32     |        3 |            30 |               28 |           0 | 2026-10-19 20:59:10 |
|            18 | 24ITR002 | 24ITL33     |        3 |            30 |               25 |           0 | 2026-10-19 20:59:10 |
|            19 | 24ITR003 | 24ITT31     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|            20 | 24ITR003 | 24ITC31     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|            21 | 24ITR003 | 24ITT32     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|            22 | 24ITR003 | 24ITC32     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|            23 | 24ITR003 | 24ITT33     |        3 |            60 |               56 |           0 | 2026-10-19 20:59:10 |
|            24 | 24ITR003 | 24MNT31     |        3 |            60 |               49 |           0 | 2026-10-19 20:59:10 |
|            25 | 24ITR003 | 24ITL31     |        3 |            30 |               28 |           0 | 2026-10-19 20:59:10 |
|            26 | 24ITR003 | 24ITL32     |        3 |            30 |               25 |           0 | 2026-10-19 20:59:10 |
|            27 | 24ITR003 | 24ITL33     |        3 |            30 |               26 |           0 | 2026-10-19 20:59:10 |
|            28 | 24CSR001 | 24CST31     |        3 |            60 |               49 |           0 | 2026-10-19 20:59:10 |
|            29 | 24CSR001 | 24CSC31     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|            30 | 24CSR001 | 24CST32     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
```

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

REFERENCES:

1. Oracle, Java SE Documentation – JDBC Overview and Architecture. https://docs.oracle.com/javase/8/docs/technotes/guides/jdbc/ .

2. Oracle, Java SE API – HttpServer. https://docs.oracle.com/en/java/javase/17/docs/api/jdk.httpserver/com/sun/net/httpserver/HttpServer.html .

3. Oracle, MySQL 8.0 Reference Manual. https://dev.mysql.com/doc/refman/8.0/en/ .

4. Oracle, MySQL Connector/J Developer Guide. https://dev.mysql.com/doc/connector-j/8.0/en/ .

5. University Grants Commission, Guidelines for Choice Based Credit System (CBCS). https://www.ugc.gov.in/pdfnews/8023719_Guidelines-for-CBCS.pdf .

6. Chart.js, Chart.js Documentation. https://www.chartjs.org/docs/latest/ .

7. Tailwind CSS, Tailwind CSS Documentation. https://tailwindcss.com/docs .

8. GeeksforGeeks, Establishing JDBC Connection in Java. https://www.geeksforgeeks.org/establishing-jdbc-connection-in-java/ .

9. YouTube, Java JDBC with MySQL Tutorial Videos. https://www.youtube.com/results?search_query=Java+JDBC+MySQL+tutorial .

10. Anthropic, Claude AI Assistant (used for project ideas and guidance). https://claude.ai/ .

GITHUB:

LINK:	**https://github.com/shankarthiclar/IA2708.git**

shankarthiclar/IA2708.git

rock/java__assignment/commit/cd8534143bf1f64498dcbec6332963cd839dccb8